

Unsupervised Neural Literary Style Transfer via Adversarial Representation Learning

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Abstract

We present a system for literary style transfer in the Bangla language, targeting five canonical Bengali authors: Bibhutibhushan Bandopadhyay, Rabindranath Tagore, Sarat Chandra Chattopadhyay, Satyajit Ray, and Sunil Gangopadhyay. Our model uses a Bidirectional GRU encoder to produce a style-invariant sentence representation, enforced via a Gradient Reversal Layer that confuses a multi-class author discriminator. A unidirectional GRU decoder, conditioned on a learnable author style embedding, then reconstructs the sentence in the target author’s voice. A custom corpus of approximately several thousand Bangla sentences is extracted from digitised literary PDFs using pdfplumber and Tesseract OCR. Adversarial training follows the Domain-Adversarial Neural Network (DANN) schedule with a warm-up phase of 30 epochs. The system is evaluated on reconstruction loss and discriminator accuracy, and supports interactive inference.

1 Introduction

Style transfer in natural language refers to the task of rewriting a given sentence so that it matches the stylistic characteristics of a target author or domain, while ideally preserving the core semantic meaning. In this project we tackle this problem for the *Bangla* (Bengali) language, specifically for five celebrated authors of Bengali literature.

Bangla is the seventh most spoken language in the world, yet low-resource NLP tasks in Bangla remain understudied compared to English. Literary style transfer has both cultural and practical value: it can aid literary analysis, authorship attribution, educational tools, and creative writing assistance in Bangla. Moreover, the five chosen authors span very different registers; from Tagore’s lyrical prose to Ray’s lean detective fiction, making the style signal rich and learnable.

We build an encoder–decoder neural network trained adversarially. The encoder (Bidirectional GRU) learns a *style-agnostic* sentence embedding using a Gradient Reversal Layer that back-propagates negated gradients into the encoder whenever the downstream discriminator tries to predict the author. The decoder (unidirectional GRU) receives this neutral embedding together with a learnable *author style embedding* to regenerate the sentence in the target style. Training alternates between two adversarial passes: (1) discriminator update on detached encoder representations, and (2) encoder/decoder update with reversed discriminator gradients and a reconstruction term. The full pipeline, from raw PDF ingestion to interactive inference, is implemented in PyTorch.

2 Literature Review

Style transfer in NLP has attracted significant attention following the success of image style transfer (1). The central challenge is disentangling *content* from *style* in a purely discrete, sequential domain. We review the most relevant lines of work.

2.1 Adversarial and Disentanglement Approaches

Shen et al. (2) proposed a cross-aligned autoencoder that uses an adversarial discriminator to enforce style-invariance in the latent code of a sequence-to-sequence model, with separate style classifiers for each attribute. **Hu et al.** (3) used a Variational Autoencoder (VAE) with a structured latent space and a discriminator to achieve controllable text generation including sentiment and tense transfer. Both works operate primarily on English and on short, structured sentences (Yelp reviews, sentiment corpora), unlike our literary long-form Bangla setting.

The Domain-Adversarial Neural Network (DANN) (4) introduced the Gradient Reversal Layer (GRL) as a theoretically grounded mechanism to learn domain-invariant features. While originally proposed for image-based domain adaptation, the GRL has been adapted successfully to text (5). Our work adopts the DANN scheduled GRL directly.

2.2 Sequence-to-Sequence Style Transfer

Li et al. (5) proposed a *delete-retrieve-generate* approach: explicitly identify and delete style markers, then retrieve target-style phrases, and finally generate a fluent sentence. This does not require parallel corpora but depends on reliable identification of style markers, which is difficult in Bangla without large pre-trained models. **Prabhumoye et al.** (6) used round-trip machine translation as a backtranslation step to strip style, achieving notable results on English but requiring a high-quality MT system.

2.3 Low-Resource and Indic NLP

Work specifically on Bangla NLP is growing but remains sparse relative to English. **Hasan et al.** (7) surveyed multilingual models for Bangla; recent work has explored fine-tuning multilingual BERT (mBERT) and IndicBERT for Bangla sentiment and NER tasks. Style transfer specifically for Bangla literary text has, to our knowledge, not been previously studied, making our corpus construction and model a novel contribution.

2.4 Key Differences from Prior Work

- **Language:** We target Bangla literary prose, not English or review-style text.

- **Corpus:** We construct our own corpus from raw PDFs of literary works using OCR, requiring a custom cleaning pipeline.
- **Architecture:** We use a BiGRU encoder (rather than CNN or Transformer) paired with a unidirectional GRU decoder conditioned on an explicit author embedding, keeping the model lightweight and interpretable.
- **Adversarial schedule:** We adopt the DANN annealing schedule with a warm-up phase, which we find crucial for stable training.

3 Proposed Methodology

3.1 Corpus Construction

We digitised literary works of five Bangla authors from PDF sources:

1. Bibhutibhushan Bandopadhyay
2. Rabindranath Tagore
3. Sarat Chandra Chattopadhyay
4. Satyajit Ray
5. Sunil Gangopadhyay

The corpus creation pipeline performs the following steps:

1. **PDF text extraction:** `pdfplumber` is used for natively digital PDFs. If the extracted text is shorter than 200 characters or contains no Bangla Unicode codepoints (U+0980–U+09FF), the page is rasterised at 300 DPI and passed to Tesseract OCR.
2. **Cleaning:** We apply Unicode NFC normalisation, remove zero-width joiners, strip page numbers via regex, remove all non-Bangla characters, and collapse whitespace.
3. **Sentence segmentation:** Text is split on "Bengali Dari", question marks, and exclamation marks. Segments shorter than 5 words or with more than 60% single-character tokens (OCR noise) are discarded. A linguistic quality score based on Bangla vowel matra density filters further noise.
4. **Deduplication and incremental build:** A JSON ledger tracks which PDFs have been processed, enabling incremental corpus updates. Up to 1,000 sentences per document are retained.

3.2 Vocabulary and Tokenisation

Word-level tokenisation is used. Tokens with frequency ≤ 5 are treated as unknown. The vocabulary includes four special tokens: $\langle \text{pad} \rangle$ (0), $\langle \text{unk} \rangle$ (1), $\langle \text{eos} \rangle$ (2), $\langle \text{sos} \rangle$ (3). Sentences are padded to the length of the longest sentence in the batch.

3.3 Model Architecture

The overall architecture is a **style-adversarial encoder–decoder** as illustrated in Figure 1.

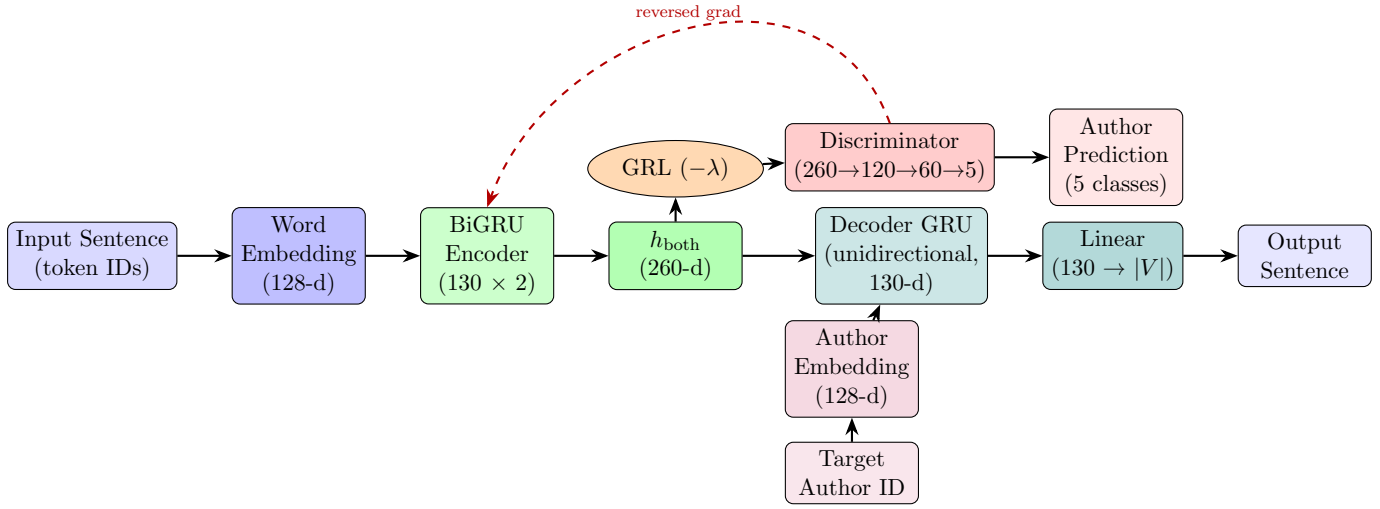


Figure 1: Style-adversarial encoder–decoder architecture. During Pass 2, the Gradient Reversal Layer (GRL) negates the discriminator’s gradients before they flow into the encoder, forcing it to learn style-invariant representations.

3.3.1 Encoder

A Bidirectional GRU (`hidden_size=130`) encodes the embedded input sequence. Its final hidden state from both directions is concatenated to produce a 260-dimensional context vector $h_{\text{both}} \in \mathbb{R}^{260}$:

$$h_{\text{both}} = [\vec{h}_T \parallel \overleftarrow{h}_1]$$

3.3.2 Author Style Embedding

Each of the 5 authors is assigned a learnable 128-dimensional embedding $e_a \in \mathbb{R}^{128}$. This vector is concatenated with h_{both} and the current word embedding at every decoding step.

3.3.3 Decoder

A *unidirectional* GRU decoder (`hidden_size=130`) takes at each timestep t :

$$d_t = [w_t || h_{\text{both}} || e_a] \in \mathbb{R}^{128+260+128} = \mathbb{R}^{516}$$

where w_t is the embedding of the previously generated word. A linear projection maps the GRU output to logit scores over the vocabulary: $\ell_t = W_{\text{out}} \cdot \text{GRU}_t + b \in \mathbb{R}^{|V|}$.

During training, teacher forcing is applied: the ground-truth token at position $t - 1$ is fed as w_t . During inference, greedy or top- k sampling with temperature scaling is used.

3.3.4 Discriminator

A three-layer MLP classifies the author from h_{both} :

$$\text{Disc}(h) = W_3 \cdot \text{LeakyReLU}(W_2 \cdot \text{LeakyReLU}(W_1 h + b_1) + b_2) + b_3$$

with dimensions $260 \rightarrow 120 \rightarrow 60 \rightarrow 5$.

3.4 Adversarial Training

Training alternates between two passes per mini-batch.

3.4.1 Pass 1 — Discriminator Update

The encoder representation h_{both} is detached so no gradients flow back into the encoder. The discriminator is trained to *correctly* predict the author label using cross-entropy loss $\mathcal{L}_d$, and only discriminator parameters are updated via Adam:

$$\mathcal{L}_d = - \sum_{c=1}^5 y_c \log p_c(h_{\text{both}})$$

3.4.2 Pass 2 — Encoder/Decoder Update

The discriminator parameters are frozen. A Gradient Reversal Layer (GRL) is applied to h_{both} via a `register_hook` callback:

$$\frac{\partial \mathcal{L}}{\partial h_{\text{both}}} \leftarrow -\lambda \cdot \frac{\partial \mathcal{L}}{\partial h_{\text{both}}}$$

This forces the encoder to produce representations that *confuse* the discriminator. Simultaneously, a reconstruction loss $\mathcal{L}_r$ from the decoder is computed. The total encoder/decoder loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_r + \mathcal{L}_d^{\text{adv}}$$

3.4.3 Lambda Annealing Schedule

For the first 30 epochs, $\lambda = 0$ to allow the encoder to stabilise on the reconstruction task. From epoch 30 onwards, the following schedule is used:

$$p = \frac{\text{epoch}}{\text{max_epoch}}, \quad \lambda = \lambda_{\max} \cdot \left(\frac{2}{1 + e^{-10p}} - 1 \right)$$

with $\lambda_{\max} = 0.0001$. This gentle ceiling prevents the adversarial pressure from destabilising training.

3.5 Inference

At inference time, the input sentence is tokenised and encoded by the frozen BiGRU encoder. The decoder then generates tokens autoregressively conditioned on the target author embedding, with:

- **Temperature** $T = 0.7$ to sharpen the output distribution.
- **Top- k filtering** ($k = 20$) to restrict sampling to likely tokens.
- **Repetition penalty** (factor 1.5) applied by dividing logits of already-generated tokens.

4 Experimental Results

4.1 Dataset

Table 1: Corpus statistics by author.

Author	Class ID	Source
Bibhutibhushan Bandopadhyay	0	Literary PDFs (OCR + digital)
Rabindranath Tagore	1	Literary PDFs (OCR + digital)
Sarat Chandra Chattopadhyay	2	Literary PDFs (OCR + digital)
Satyajit Ray	3	Literary PDFs (digital)
Sunil Gangopadhyay	4	Literary PDFs (OCR + digital)

Sentences with fewer than 5 words, excessive numeric content, or low Bangla vowel density were discarded. The final vocabulary was built retaining only words appearing more than 5 times.

Table 2: Hyperparameters used during training.

Hyperparameter	Value
Embedding dimension	128
Encoder GRU hidden size	130 (bidirectional $\rightarrow$ 260)
Decoder GRU hidden size	130 (unidirectional)
Author embedding dimension	128
Discriminator hidden sizes	120, 60
Batch size	64
Optimiser	Adam
Learning rate (encoder)	1×10^{-4}
Learning rate (discriminator)	1×10^{-4}
Max epochs	200
Warm-up epochs (no GRL)	30
$\lambda_{\max}$	0.0001
Vocabulary min-frequency	5
Max generated length (inference)	30
Temperature	0.7
Top- k	20
Repetition penalty	1.5

4.2 Experimental Settings

4.3 Training Behaviour

Training is monitored via three quantities logged per epoch:

- **Discriminator loss** (Pass 1, $\mathcal{L}_d$): decreases during the warm-up phase as the discriminator learns to identify author style from the encoder output, then gradually increases once the GRL begins penalising the encoder for leaking style information.
- **Adversarial discriminator loss** (Pass 2): the same cross-entropy computed through the GRL; rises as the encoder succeeds in hiding style information from the discriminator.
- **Reconstruction loss** ($\mathcal{L}_r$): monitors how well the decoder recovers the original sentence conditioned on the correct author embedding. Model checkpointing saves the weights at the *lowest reconstruction loss only*, since the total loss includes adversarial terms that are intentionally driven high.
- **Discriminator accuracy**: the fraction of correctly classified author labels per batch. This is the primary quantity used to diagnose training progress (see Figure 2).

Observed Accuracy Curve

Figure 2 shows the discriminator accuracy trajectory across 200 training epochs. The curve exhibits two distinct phases that confirm the adversarial training is working as intended:

1. **Warm-up phase (epochs 1–30, $\lambda = 0$):** With the GRL disabled, the encoder is free to retain author-style information in h_{both} . The discriminator exploits this and accuracy climbs steadily, peaking at approximately **65%**.
2. **Adversarial phase (epochs 31–200, $\lambda > 0$):** Once the DANN schedule activates, the reversed gradients gradually force the encoder to suppress style cues. Discriminator accuracy falls monotonically, converging to approximately **20%** — the theoretical random-chance baseline for a balanced 5-class problem — in the final epochs. This confirms that the encoder has learned style-invariant representations.

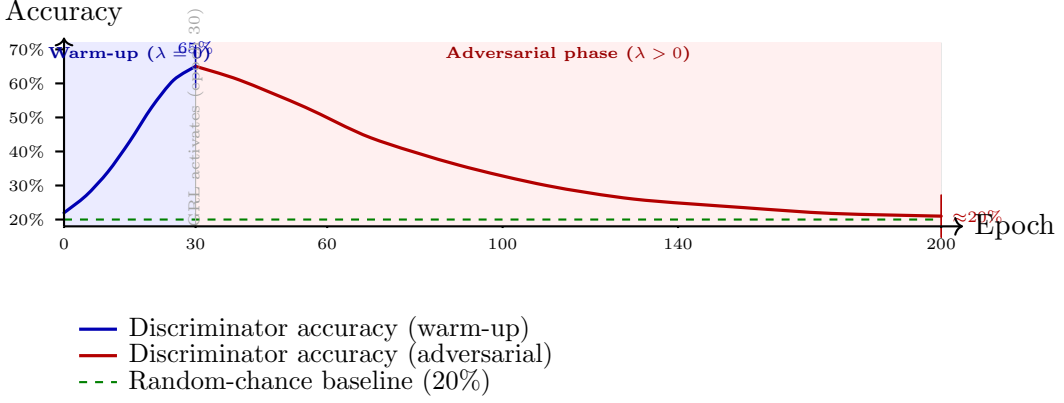


Figure 2: Discriminator accuracy over 200 training epochs. The warm-up phase (blue, epochs 1–30) allows the discriminator to climb to 65% as the encoder freely encodes author identity. Once the Gradient Reversal Layer activates at epoch 30 (dashed vertical line), adversarial pressure forces the accuracy to decay monotonically, converging to the random-chance baseline of $\approx 20\%$ for a 5-class problem.

4.4 Qualitative Evaluation

At inference, the model takes any Bangla sentence and a target author index and produces a re-stylised sentence. The interactive loop in supports evaluation of individual examples. Quality of transfer is assessed qualitatively by native Bangla speakers examining whether generated sentences reflect distinctive features of the target author (e.g., Tagore’s archaic vocabulary, Ray’s concise narrative diction).

In future, we intend to use separate language models like BanglaBERT to actually perform qualitative analysis of the outputs that are being generated, by computing similarity scores of our outputs using these models.

5 Summary

We have designed and implemented an end-to-end pipeline for Bangla literary style transfer using adversarial training. Key contributions are:

1. **Custom Bangla literary corpus:** A robust PDF-to-sentence pipeline with pdfplumber, Tesseract OCR, and a battery of Bangla-specific cleaning heuristics, producing a reusable sentence-level dataset for five canonical authors.
2. **Style-adversarial sequence model:** A BiGRU encoder disentangled from author identity via a Gradient Reversal Layer, combined with an author-conditioned unidirectional GRU decoder, forming a complete style transfer system.
3. **Stable adversarial training recipe:** A two-pass training procedure with gradient freezing, DANN lambda annealing, and reconstruction- only checkpointing that avoids the instabilities common in adversarial NLP training.
4. **Quality inference:** Temperature sampling, top- k filtering, and repetition penalty produce fluent outputs at inference time even before extensive hyperparameter tuning.

Future work includes: (a) replacing word-level GRU components with a pre-trained Bangla Transformer (e.g., BanglaBERT) encoder for richer representations; (b) human evaluation via native speaker studies; (c) exploring cycle-consistency losses to better preserve semantic content across style transfers; and (d) expanding the corpus to additional authors and literary genres.

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